Infographic Presentation (Prep)
Final Infographic Due: November 24

1. **Project title**
   Equitable Grade Weighting Matrix

2. **Group member names**
   Rachel Eglash, Peter Kudelin, Sena Lee

3. **Summary, including:**
   A. **the research question and linear system or linear algebra-related mathematical model or theory that will be used to answer this question**

      Do different grade weighting methods produce more equitable outcomes across different demographic groups?
         Additional research questions:
            Which specific demographic groups does this apply to (gender, socioeconomic status, disability status, race, etc.)?
            How does including homework and "employability," (from Griffin's dataset, or a combination of participation, attendance, and behavior from the UCI dataset) change the final grades for students in different demographic groups?
            What is the magnitude of the equity gap (the difference in the mean grades) in different demographic groups under different grade weighting methods?
            Which demographic groups benefit the most from traditional grading that includes non-academic factors, and which are most disadvantaged?

   B. **the name(s) of the linear algebra technique(s) from class used**

      Weighted linear combinations and matrix-vector multiplication
         Student grades can be modeled as weighted linear combinations of multiple grade components. The different grade weighting methods represent different weights of vectors applied to the same group of students with their given grades (represented in a matrix). By calculating final grades using multiple different grade weighting methods using matrix-vector multiplication and comparing the equity gaps between different demographic groups, we can determine which grading method is most equitable (while still maintaining academic rigor requirements).

C. **the motivation behind the project: why do you care about his application? Why should the reader care about this application?**

As students, we all know that the grading system that our professors use can have a large impact on our academic trajectory, including things like college admission and future opportunities. However, some grading methods may disadvantage certain demographic groups, inadvertently or not, who face many systemic barriers outside the classroom.
Understanding which grading methods minimize demographic disparities is crucial for creating fair and equitable educational systems. This research provides data-driven insights that can help those of us who want to become educators, to use grading practices that accurately reflect student's learning rather than perpetuating existing inequalities.

D. **The field linear algebra techniques are applied to in this project (such as physics, ecology, chemistry, etc.)**

Education

E. **Definitions of any specialized terms or summaries of concepts related to the field someone from this class may not be familiar with. The only undefined terms may be terms from this class and its prerequisites.**

Equity – Fairness in educational outcomes, where students from different demographic backgrounds have equal opportunity to demonstrate their knowledge and achieve success; in this project, equity refers to minimizing systematic differences in grades between demographic groups that result from factors unrelated to academic mastery
Grade weighting method – A system that assigns different percentages or weights to various components of a student's performance (such as assessments, homework, and employability) to calculate a final grade; different weighting methods prioritize different aspects of student performance
Demographic groups – Categories of students based on shared characteristics such as race, gender, socioeconomic status, or disability status; these groups are used to analyze whether grading methods produce systematically different outcomes for different populations
Gender – Demographic group; for the purposes of this project, the determination of a student's gender is binary (male or female)
Disability – Demographic group; for the purposes of this project, a student's disability status is determined by whether or not they have an IEP or 504 plan
IEP – Individualized Education Plan; a legally binding document that outlines specialized instruction and services for students with disabilities; used as a measure for disability status in this project
504 plan – A plan that provides accommodations and modifications for students with disabilities; used as a measure for disability status in this project
SES – Socioeconomic Status; demographic group; for the purposes of this project, a student's SES is determined by whether or not they receive FRL
FRL – Free or Reduced Lunch; a federal program that provides meal assistance to students from low-income families; used as a measure for SES in this project
Race – Demographic group; for the purposes of this project, a student's race was either White, Black, Hispanic, Asian, or Other
Assessment score – A measure of a student's completion and performance on assignments completed outside of class
Homework score – A measure of a student's performance on tests and quizzes
Employability score – A measure of non-academic factors such as participation, attendance, and behavior

4. **Answer to the research question and solution of the linear system (or linear algebra-related mathematical model or claim)**
    A. **Start with the research question and any additional context for the example shown. This could read like a word problem, without symbolic math or math-specific vocabulary. In other words, a student not in a math class should be able to read and understand the context and question.**

    To determine if different grade weighting methods produce more equitable outcomes across different demographic groups, we will start by determining the final grade using two different grade weighting methods (assessment-only and traditional) for 5 students which included one student from each race (White, Black, Hispanic, Asian, and Other).
    The final grade for a given student is determined by a combination of their assessment score, homework score, and employability score. Each different grade weighting method uses a different combination or percentage from these three scores to determine the final grade.
        For the assessment-only grade weighting method, 100% of the final grade comes from the assessment score (the homework score and employability score are not taken into account).
        For the traditional grade weighting method, 70% of the final grade comes from the assessment score, 20% of the final grade comes from the homework score, and 10% of the final grade comes from the employability score.
    We will compare the difference in these final grades from different grade weighting methods to determine if we see a significant change for certain races more than others (or another demographic group) when using different grade weighting methods.
    By computing final grades under multiple weighting methods and calculating the resulting equity gaps between different demographic groups, we can determine which method minimizes disparities. The method that produces the smallest equity gaps while maintaining academic standards is the most equitable method.

B. Define all relevant variables used in the computation shown to answer the research question, including units (if applicable).

$m =$ total number of students
For student i $(1 \leq i \leq m)$:
$a_i =$ assessment score
$h_i =$ homework score
$e_i =$ employability score
$g_i =$ final grade
where $a_i, h_i, e_i, g_i \in [0,100]$
For each grade weighting method:
$w_a =$ weight for assessment scores
$w_h =$ weight for homework scores
$w_e =$ weight for employability scores
where $w_a, w_h, w_e \in [0,1] : w_a + w_h + w_e = 1$
$A = m \times 3$ matrix containing the three scores for each student
$w = 3 \times 1$ vector containing the weights for the three scores for each student
$g = m \times 1$ column vector containing the final grade for each student

C. **Explain the steps require to "translate" the research question into an equation or computation. In other words, explain how someone could use linear algebra to answer the question? Try to teach your peers how to answer similar questions using linear algebra.**

The general form of the linear system that we will be solving is $Ax = b$, or more specifically for our research equation, $Aw = g$, where:

$$A = \begin{bmatrix} a_1 & h_1 & e_1 \\ \vdots & \vdots & \vdots \\ a_i & h_i & e_i \\ \vdots & \vdots & \vdots \\ a_m & h_m & e_m \end{bmatrix}, \qquad w = \begin{bmatrix} w_a \\ w_h \\ w_e \end{bmatrix}, \qquad g = \begin{bmatrix} g_1 \\ \vdots \\ g_i \\ \vdots \\ g_m \end{bmatrix}$$

Therefore, for a given grade weighting method, where $w_a + w_h + w_e = 1$, using weighted linear combinations and matrix-vector multiplication, the final grade for each student is calculated and is given by the equation:

$$\begin{bmatrix} a_1 & h_1 & e_1 \\ \vdots & \vdots & \vdots \\ a_i & h_i & e_i \\ \vdots & \vdots & \vdots \\ a_m & h_m & e_m \end{bmatrix} \begin{bmatrix} w_a \\ w_h \\ w_e \end{bmatrix} = \begin{bmatrix} g_1 \\ \vdots \\ g_i \\ \vdots \\ g_m \end{bmatrix}$$

For a given student i, their final grade for a given grade weighting method is:

$$g_i = [a_i \quad h_i \quad e_i] \cdot \begin{bmatrix} w_a \\ w_h \\ w_e \end{bmatrix} = w_a \cdot a_i + w_h \cdot h_i + w_e \cdot e_i$$

In different grade weighting methods, the weights take assessment score, homework score, and employability score into account, where the weights for the three score must add to 1.

In the assessment-only grade weighting method, which only takes tests into account, the weight for the assessment score is 1, the weight for the homework score is 0, and the weight for the employability score is 0.

In the traditional grade weighting method, the weight for the assessment score is 0.7, the weight for the homework score is 0.2, and the weight for the employability score is 0.1.

D. **You don't need to show all the computations, especially if they are lengthy. Label where the linear algebra techniques you named were used in this computation.**

In Trial 1, we used a random number generator to randomly select a small sample of 5 students. Using matrix vector multiplication, we calculated the final grades for these 5 students using both the traditional grade weighting method and the assessment-only grade weighting method. We repeated this process in Trial 2 and Trial 3, with another randomly selected sample of 5 students (see Figure 1). These computations are summarized in Figure 2.

Additionally, we created a computer program in C++ to calculate the final grades for all students using the assessment-only grade weighting method and the traditional grade weighting method.

This program can be found at this GitHub link:
https://github.com/r-eglash/Math2B-Project-Fall-2025/blob/main/Program_and_Raw_Data/main.cpp
The raw data for the assessment-only weighting method and the traditional weighting method can be found here:
https://github.com/r-eglash/Math2B-Project-Fall-2025/blob/main/Program_and_Raw_Data/Raw_Data/raw_data_full.csv

E. **Write (in words) a conclusion interpreting the result of the computation, including any relevant units. Make sure to answer the research question you began with.**

By computing final grades under multiple weighting methods and calculating the resulting equity gaps between different demographic groups, the method minimizes disparities can be determined. The method that produces the smallest equity gaps while maintaining academic standards is the most equitable method.

When switching from the assessment-only grade weighting method to the traditional grade weighting method, we found significant equity gaps across all demographic categories extended.

- Students without disabilities gained an average of 5.2 points, whereas students with disabilities gained an average of 2.8 points. This creates an equity gap of 2.4 points.
- White students gained an average of 5.8 points, Asian students gained an average of 5.3 points, Hispanic students gained an average of 4.1 points, and Black students gained an average of 3.2 points. This creates an equity gap of 2.6 points between White and Black students.
- Students not receiving FRL gained an average of 6.1 points, whereas students receiving FRL gained an average of 3.7 points. This creates an equity gap of 2.4 points.
- Female students gained an average of 5.4 points, whereas male students gained an average of 4.8 points. This creates an equity gap of 0.6 points (the smallest disparity).

Based on the analysis, the assessment-only grading method produces smaller equity gaps across all demographic categories examined.

There are several important limitations, both based on demographics and the model scope:

Demographic Category Limitations:

- Disability determination: This model only counts students with formal IEP or 504 plans. Many students have undiagnosed disabilities (such as neurodivergence, ADHD, dyslexia, anxiety disorders) that affect their homework completion but aren't formally documented. This means this model likely underestimates the impact on students with disabilities.
- Race determination: The dataset uses single-race categories, which doesn't capture the experience of biracial or multiracial students. Additionally, different subsets of populations (like different Asian backgrounds, such as East Asian, Southeast Asian, South Asian) have vastly different educational experiences, but are grouped together as "Asian" in the data.
- SES determination: Based solely on whether students receive free or reduced lunch (FRL). This doesn't capture students just above the eligibility cutoff who still face financial hardship, or students experiencing temporary housing instability.
- Gender determination: The school's gender identity for a student may differ from the student's own gender identity. The binary categorization doesn't include nonbinary students or account for transgender students.

Model Scope Limitations:

This model measures one aspect of equity, which is differences in grade outcomes between demographic groups. However, it doesn't capture:

- Whether the assessments themselves are culturally biased
- Whether grades actually reflect learning
- Quality of instruction received by different groups
- Out-of-school factors affecting performance

F. **Include a graph of the solution of the linear system used to answer your research question. If you didn't solve a linear system, provide a diagram or other graph that is related to the work you showed to answer the research question and that helps the reader interpret the context and solution. If using a graph, label the axes (including units). Provide a brief captain describing what the graph shows. If you show more than one IVP on your infographic, make sure to clarify which solution is in the graph.**

The results from our data for the final grades for all students using two different grade weighting methods, assessment-only and traditional, are summarized in the bar graph in Figure 4. Each graph shows the average (mean) change in score (when changing from the assessment-only grade weighting method to the traditional grade weighting method) for each grade weighting method for each category in a given demographic (disability, race, socioeconomic status, gender).

Our results are also summarized in the pivot tables in Figure 5.

Figure 5a shows a pivot table for disability, where "yes" means the student has an IEP or 504 plan (and is considered "disabled" for the purpose of this project), and "no" means no IEP or 504 plan (and is considered "not disabled" for the purpose of this project). See the results section for a discussion on undiagnosed disabilities.

Figure 5b shows a pivot table for race, where a student is identified as "White," "Black," "Hispanic," "Asian," or "other." See the results section for a discussion on biracial students (or students identifying as more than one category) and different subsets of populations (like different Asian backgrounds).

Figure 5c shows a pivot table for FRL (socioeconomic status). "Yes" means the student receives a free or reduced lunch (and is considered to have a lower socioeconomic status for the purpose of this project) and "no" means the student does not receive a free or reduced lunch (and is considered to have a higher socioeconomic status for the purpose of this project).

Figure 5d shows a pivot table for gender, where the school is identifying the student as either "male" or "female." See results section for a discussion on the school's gender identity for the student may differ from the student's gender identity (or may identify as nonbinary).

5. **Cite all sources at the bottom of the infographic.**

Cortez, P. and A. M. Gonçalves Silva. "Using data mining to predict secondary school student performance." (2008). https://www.semanticscholar.org/paper/Using-data-mining-to-predict-secondary-school-Cortez-Silva/61d468d5254730bbecf822c6b60d7d6595d9889c

Griffin, Robert Thomas, "Grading and equity: Inflation/deflation based on race, gender, socio-economic and disability statuses when homework and employability scores are included" (2020). *Dissertations and Theses @ UNI*. 1045. https://scholarworks.uni.edu/etd/1045

Figure 1: Calculations for three samples of students using two grade weighting method

Trial 1

Students 540, 245, 708, 530, 586

#	a	h	e
540	75.08	99.05	90.14
245	55.97	57.31	88.64
708	44.5	47.08	21.35
530	77.87	97.73	96.71
586	85.37	100	100

Assessment - only $\begin{bmatrix} 75.08 \\ 55.97 \\ 44.5 \\ 77.87 \\ 85.37 \end{bmatrix}$ Traditional $\begin{bmatrix} 81.38 \\ 59.505 \\ 42.701 \\ 83.726 \\ 89.759 \end{bmatrix}$

Assessment-only method

$$\begin{bmatrix} 75.08 & 99.05 & 90.14 \\ 55.97 & 57.31 & 88.64 \\ 44.5 & 47.08 & 21.35 \\ 77.87 & 97.73 & 96.71 \\ 85.37 & 100 & 100 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 75.08 \\ 55.97 \\ 44.5 \\ 77.87 \\ 85.37 \end{bmatrix}$$

Traditional Method

$$\begin{bmatrix} 75.08 & 99.05 & 90.14 \\ 55.97 & 57.31 & 88.64 \\ 44.5 & 47.08 & 21.35 \\ 77.87 & 97.73 & 96.71 \\ 85.37 & 100 & 100 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.2 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 52.556 + 19.81 + 9.014 \\ 39.179 + 11.462 + 8.864 \\ 31.15 + 9.416 + 2.135 \\ 54.509 + 19.546 + 9.671 \\ 59.759 + 20 + 10 \end{bmatrix} = \begin{bmatrix} 81.38 \\ 59.505 \\ 42.701 \\ 83.726 \\ 89.759 \end{bmatrix}$$

Trial 2

Students 450, 135, 280, 20, 11

#	a	h	e
450	50.91	77.54	64.94
135	64.1	97.22	100
280	56.28	21.87	49.36
20	60.31	78.81	88.98
11	86.53	100	88.79

Assessment - only $\begin{bmatrix} 50.91 \\ 64.1 \\ 56.28 \\ 60.31 \\ 86.53 \end{bmatrix}$ Traditional $\begin{bmatrix} 57.639 \\ 74.314 \\ 48.706 \\ 66.877 \\ 89.45 \end{bmatrix}$

Assessment-only method

$$\begin{bmatrix} 50.91 & 77.54 & 64.94 \\ 64.1 & 97.22 & 100 \\ 56.28 & 21.87 & 49.36 \\ 60.31 & 78.81 & 88.98 \\ 86.53 & 100 & 88.79 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 50.91 \\ 64.1 \\ 56.28 \\ 60.31 \\ 86.53 \end{bmatrix}$$

Traditional Method

$$\begin{bmatrix} 50.91 & 77.54 & 64.94 \\ 64.1 & 97.22 & 100 \\ 56.28 & 21.87 & 49.36 \\ 60.31 & 78.81 & 88.98 \\ 86.53 & 100 & 88.79 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.2 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 35.637 + 15.508 + 6.494 \\ 44.87 + 19.444 + 10 \\ 39.396 + 4.374 + 4.936 \\ 42.217 + 15.762 + 8.898 \\ 60.571 + 20 + 8.879 \end{bmatrix} = \begin{bmatrix} 57.639 \\ 74.314 \\ 48.706 \\ 66.877 \\ 89.45 \end{bmatrix}$$

Trial 3

Students 755, 704, 6, 281, 682

#	a	h	e
755	72.61	20.55	106.41
704	70.8	21.25	22.7
6	57.6	30.5	60.72
281	75.67	104.87	96.84
682	85	83	79.18

Assessment - only $\begin{bmatrix} 72.61 \\ 70.8 \\ 57.6 \\ 75.67 \\ 85 \end{bmatrix}$ Traditional $\begin{bmatrix} 65.578 \\ 56.08 \\ 58.564 \\ 83.627 \\ 84.018 \end{bmatrix}$

Assessment-only method

$$\begin{bmatrix} 72.61 & 20.55 & 106.41 \\ 70.8 & 21.25 & 22.7 \\ 57.6 & 30.5 & 60.72 \\ 75.67 & 104.87 & 96.84 \\ 85 & 83 & 79.18 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 72.61 \\ 70.8 \\ 57.6 \\ 75.67 \\ 85 \end{bmatrix}$$

Traditional Method

$$\begin{bmatrix} 72.61 & 20.55 & 106.41 \\ 70.8 & 21.25 & 22.7 \\ 57.6 & 30.5 & 60.72 \\ 75.67 & 104.87 & 96.84 \\ 85 & 83 & 79.18 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.2 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 50.827 + 4.11 + 10.641 \\ 49.56 + 4.25 + 2.27 \\ 40.32 + 6.1 + 12.144 \\ 52.969 + 20.974 + 9.684 \\ 59.5 + 16.6 + 7.918 \end{bmatrix} = \begin{bmatrix} 65.578 \\ 56.08 \\ 58.564 \\ 83.627 \\ 84.018 \end{bmatrix}$$

Figure 2: Summary of calculations from Figure 1

Trial 1

Student #	Final Grade Using Different Testing Methods		Demographic Information			
	Assessment-Only Method	Traditional Method	Disability	Race	FRL	Gender
540	75.08	81.38	No	White	Yes	Female
245	55.97	59.505	No	Black	Yes	Male
708	44.5	42.701	Yes	Black	Yes	Female
530	77.87	83.726	No	Hispanic	Yes	Male
586	85.37	89.759	No	White	Yes	Female

Trial 2

Student #	Final Grade Using Different Testing Methods		Demographic Information			
	Assessment-Only Method	Traditional Method	Disability	Race	FRL	Gender
450	50.91	57.639	No	Black	Yes	Female
135	64.1	74.314	No	White	No	Female
280	56.28	48.706	No	Black	Yes	Female
20	60.31	66.877	No	White	No	Female
11	86.53	89.45	No	White	Yes	Female

Trial 3

Student #	Final Grade Using Different Testing Methods		Demographic Information			
	Assessment-Only Method	Traditional Method	Disability	Race	FRL	Gender
755	72.61	65.578	No	Black	Yes	Male
704	70.8	56.08	No	Black	Yes	Male
6	57.6	58.564	No	White	Yes	Female
281	75.67	83.627	No	White	Yes	Female
682	85	84.018	No	White	Yes	Male

Figure 3: Graph and analysis of data from Figure 2

NEED TO DO

Figure 4: Mean Change in Score by Demographic

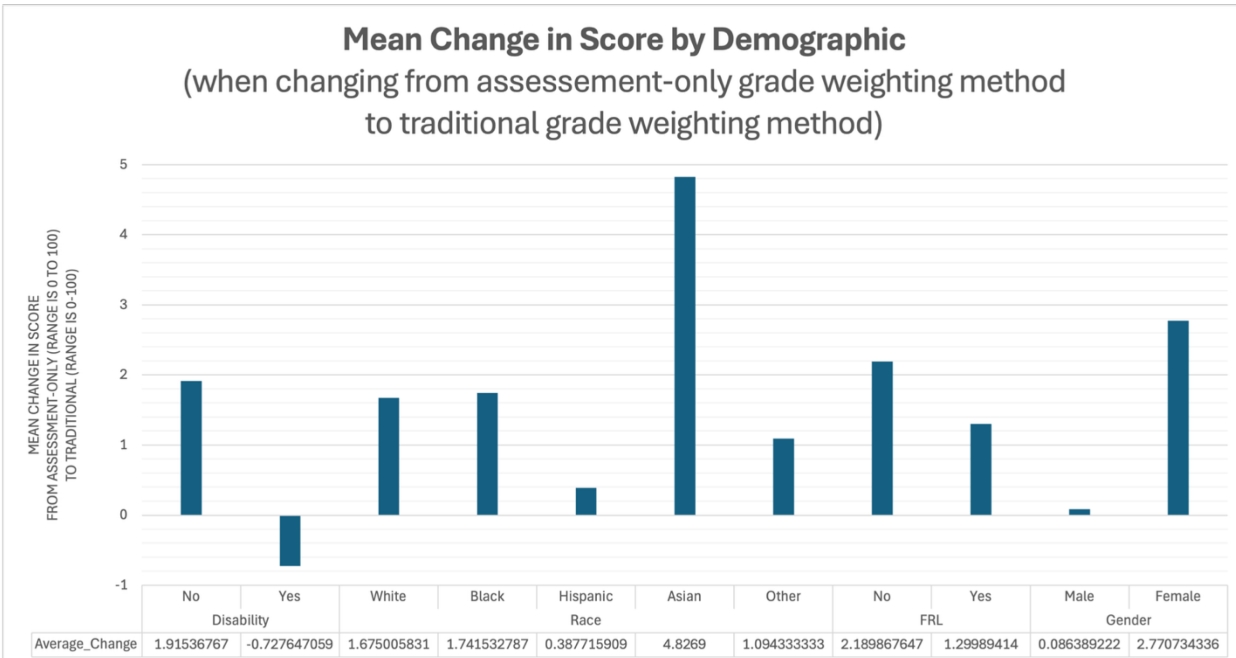

Figure 5: Pivot Tables by Demographic

Figure 5a: Pivot Table for Disability

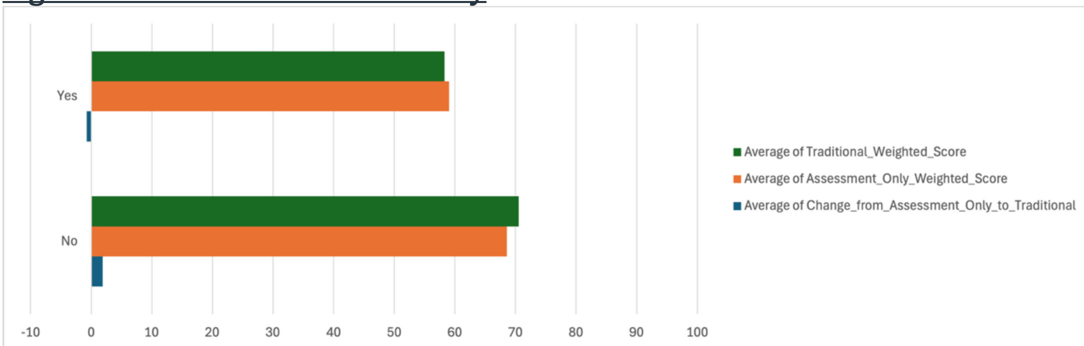

Figure 5b: Pivot Table for Race

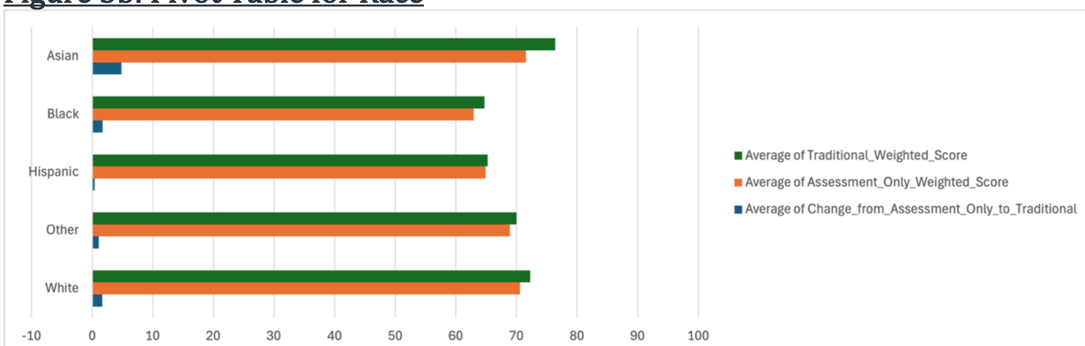

Figure 5c: Pivot Table for FRL (Socioeconomic Status)

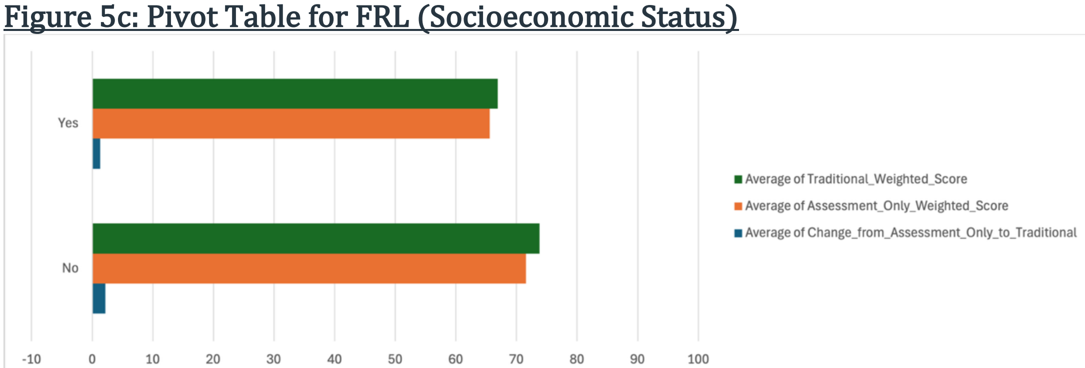

Figure 5d: Pivot Table for Gender